

UNIT 8

MIXTURES AND SOLUTIONS

Learning objectives:

- Define mixtures and solutions and differentiate between them using examples from daily life.
- Identify the components of a solution (solute and solvent) and describe their roles in forming a solution.
- Explain methods of separating mixtures, such as handpicking, filtration, and sieving, with examples.
- Recognize the importance of mixtures and solutions in everyday activities like cooking, cleaning, and medicine.

In our daily lives, we often mix things together, like sugar in water or fruits in a salad. But have you ever wondered what happens when things are mixed? In this chapter, we will learn about mixtures and solutions, how they are made, and why they are important.

WHAT IS A MIXTURE?

Mixture is formed when two or more things are combined without changing their basic properties. For example, when you mix rice and beans, you can still see and separate each part.



Types of Mixtures

- **Homogeneous Mixture**

A homogeneous mixture looks the same throughout.

Examples:

- o Salt in water
- o Milk

- **Heterogeneous Mixture**

A heterogeneous mixture has visible parts.

Examples:

- o Fruit chaat
- o Salad

Examples of Mixtures:

- Sand and water
- Cereal and milk
- Coins in a jar



WHAT IS A SOLUTION?

A **solution** is a special type of mixture where one substance completely dissolves in another. The substances in a solution cannot be separated easily.



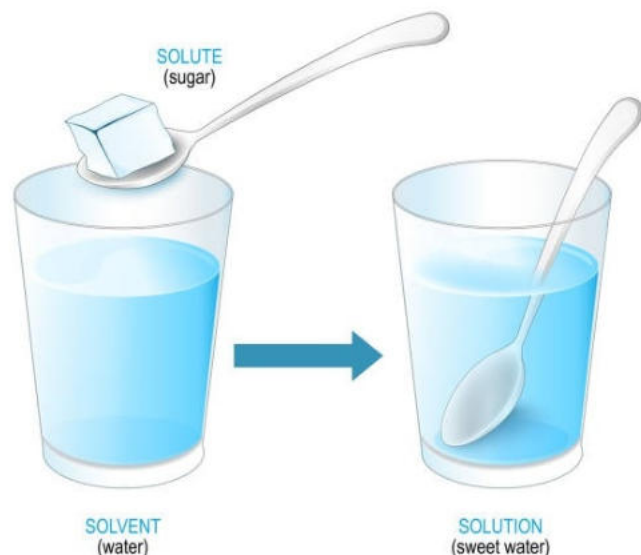
Parts of a Solution

- **Solvent:** The substance in which another substance dissolves.
(Example: Water)
- **Solute:** The substance that dissolves in the solvent.
(Example: Salt)

When salt dissolves in water, salt is the solute, and water is the solvent. Together, they form a saltwater solution.

Examples of Solutions:

- Saltwater
- Sugar dissolved in tea
- Lemonade



HOW TO SEPARATE MIXTURES

We can separate mixtures in different ways depending on their type:

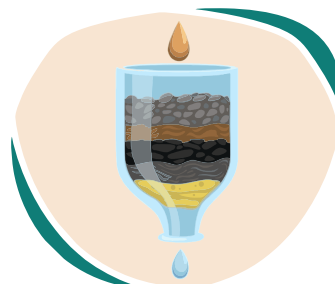
1. Handpicking

- Used to separate large, visible pieces (e.g., separating stones from rice).



2. Filtration

- Used to separate solids from liquids (e.g., sand from water).



3. Sieving

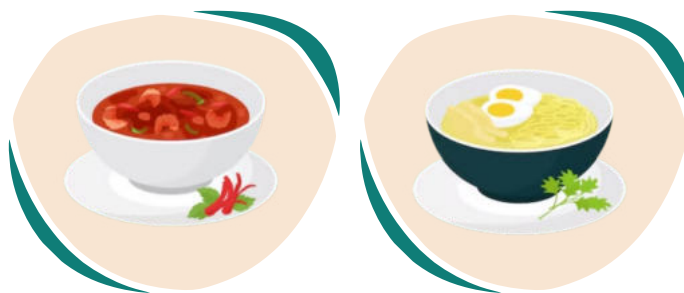
- Used to separate small particles (e.g., flour and sugar).



IMPORTANCE OF MIXTURES AND SOLUTIONS

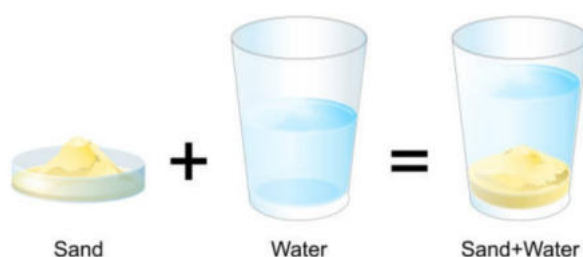
Mixtures and solutions are very useful in our daily lives:

- **Cooking:** Mixing ingredients to prepare food.
- **Medicine:** Solutions like syrups help us when we are sick.
- **Cleaning:** Soap and water make a solution that helps clean dirt.



Activity 1: Identify Mixtures Around You

- Look around your kitchen or home and write down five mixtures you can find.



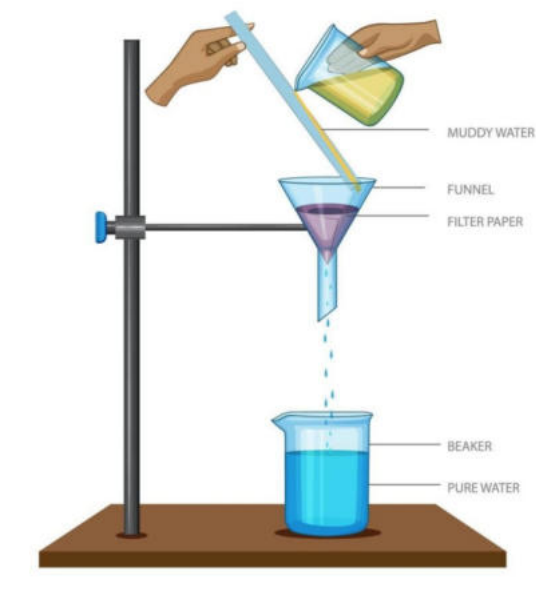
Activity 2: Making a Solution

- Take a glass of water.
- Add a teaspoon of sugar and stir.
- Observe how the sugar dissolves completely.



Activity 3: Separating a Mixture

- Mix sand and water in a bowl.
- Use a filter to separate the sand from the water.





Fill in the blanks.

1. A _____ is made when two or more substances are combined.
2. _____ is a process used to separate sand from water.
3. A solution is a special type of _____ .
4. The _____ in a solution is the substance that dissolves.
5. Saltwater is an example of a _____ .
6. A mixture of stones and rice can be separated by _____ .





Write "T" for True or "F" for False

1. A solution is always a homogeneous mixture.

2. Sugar dissolved in water forms a mixture.

3. Filtration is used to separate salt from water.

4. A solution contains a solute and a solvent.

5. Mixtures cannot be separated by handpicking.

6. Mixing flour and sugar creates a solution.



Choose the correct answer.

1. What is the part of a solution that dissolves called?

- Solvent
 - Solute
 - Filter
 - Mixture
-

2. Which method is used to separate sand from water?

- Handpicking
 - Filtration
 - Boiling
 - Sieving
-

3. What is an example of a solution?

- Rice and beans
 - Lemonade
 - Sand and water
 - Cereal and milk
-



4. Which of the following is NOT a mixture?

- Fruit salad
 - Saltwater
 - Pure gold
 - Sugar in tea
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5. Which method is best for separating stones from rice?

- Filtration
 - Handpicking
 - Sieving
 - Boiling
-

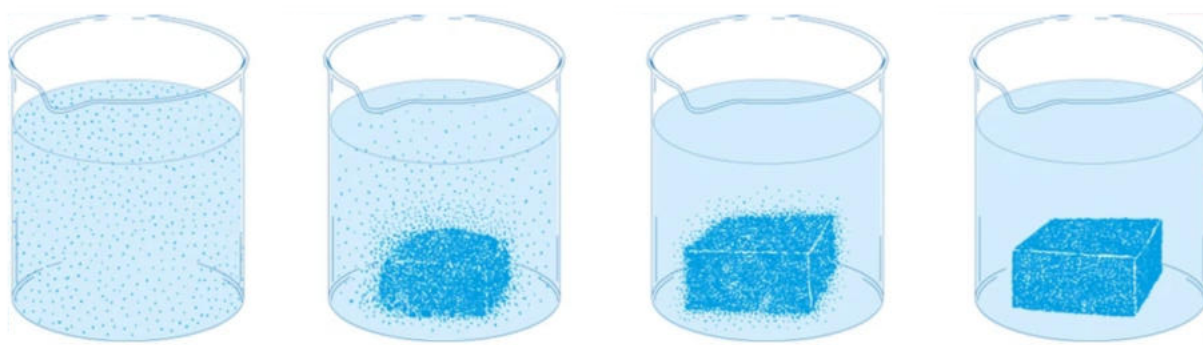
6. What is the main purpose of filtration?

- To mix two liquids
 - To separate solids from liquids
 - To make a solution
 - To dissolve substances
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Answer the following questions.

- 1. What is the difference between a mixture and a solution?**
- 2. Define homogeneous and heterogeneous mixtures.**
- 3. Give two examples of mixtures and two examples of solutions.**
- 4. What are the roles of solute and solvent in a solution?**
- 5. Name three methods of separating mixtures and explain when to use each.**
- 6. Why are mixtures and solutions important in our daily lives?**



Activity 1: Identifying Mixtures and Solutions

- Look around your house or classroom.
- Make a list of three mixtures and two solutions you see.
- Write down how you think each mixture can be separated.

Activity 2: Making a Solution

- Take a glass of water.
- Add a spoon of sugar and stir.
- Write down your observations: What happens to the sugar?
- Now, try adding sand to the same water. Can you stir it and dissolve the sand?

Activity 3: Separating Mixtures

- Mix small stones and flour in a bowl.
- Try using a sieve to separate them.
- Write down what you observe after sieving.

Activity 4: Drawing and Labelling

- Draw a diagram of a filtration process.
- Label the filter paper, mixture, and the separated parts.



Activity 5: Class Experiment

- The teacher will prepare three mixtures:
 1. Sand and water
 2. Salt and water
 3. Rice and beans
- Work in groups to figure out the best way to separate each mixture. Write your methods and results.

